

DSC 102 Spring 2026 Midterm Exam - Version C

Solution key for authorized study/review use. Arithmetic is shown where it affects the answer.

Number Representations

Q1(a): 8 bits

$2^7 = 128$ is too small; $2^8 = 256$ covers 245 distinct event types.

Q1(b): 11000110

$198 = 128 + 64 + 4 + 2$, so the 8-bit pattern is 11000110.

Q2: 0xCC

$204 = 12 \cdot 16 + 12$, and $12 = C$.

Q3(a): 15

For 5 exponent bits, IEEE-style bias = $2^{(5-1)} - 1 = 15$.

Q3(b): float16

float16 has 10 mantissa bits versus 7 in bfloat16, so it preserves more precision.

Q4: 11000011

61 is 00111101; invert and add 1 to encode -61.

Q5: -36

$110 + 110 = 220$. In 8-bit two's complement, $220 \bmod 256$ represents $220 - 256 = -36$.

Computer Organization / Architecture

Q1.1: 2.0

Average CPI = $0.60 \cdot 1 + 0.20 \cdot 5 + 0.20 \cdot 2 = 2.0$.

Q1.2: 1.0×10^{10} cycles

TC = $2.0 \text{ CPI} \cdot 5 \times 10^9 \text{ instructions}$.

Q1.3: 2 seconds

ET = $1.0 \times 10^{10} \text{ cycles} / 5 \times 10^9 \text{ cycles/s}$.

Q2.1: 3×10^9 cycles

Every instruction has 1 base cycle: $3 \times 10^9 \cdot 1$.

Q2.2: 7.2×10^{10} cycles

Memory ops = $0.40 \cdot 3 \times 10^9 = 1.2 \times 10^9$; stalls = $1.2 \times 10^9 \cdot 60$.

Q2.3: 25 seconds

Total cycles = $3 \times 10^9 + 7.2 \times 10^{10} = 7.5 \times 10^{10}$; divide by $3 \times 10^9 \text{ cycles/s}$.

Q3(a): 2.22 GHz

The clock period is limited by the slowest stage, 0.45 ns. Frequency = $1 / 0.45 \text{ ns}$.

Q3(b): 1.8 seconds

With one instruction per cycle, time = $4 \times 10^9 \cdot 0.45 \text{ ns}$.

Operating Systems

Q1(a): Page offset = 6 bits; VPN = 9 bits

A 64-byte page needs $\log_2(64) = 6$ offset bits, leaving 9 bits for the VPN.

Q1(b): 10 KB

Each process has 2^9 pages * 4 bytes/PTE = 2048 bytes. For 5 processes: 10,240 bytes = 10 KB.

Q1(c): 0x5FA

Virtual address 0x5E3A has 6-bit offset 0x3A. Physical address = $(0x17 \ll 6) + 0x3A$.

Q2(a): 20 processes

A creates 2 processes. B creates 2 children; those children expand to 8 processes inside C and D, plus 2 B-parents. E then doubles 10 to 20.

Q2(b): 19 children

Total children created = final process count - original = 20 - 1.

Q3(a): lock after L2; unlock after L5

The critical section is the full read-modify-write sequence on orders_fulfilled.

Q3(b): 1000

With correct locking, all 5×200 increments are preserved.

Q3(c): 76

Both threads can read 75, compute 76, and write 76; one increment is lost.

Q4.1: 28.75 ms

FIFO waits are $P1=0, P2=25, P3=40, P4=50$; average = $115/4$.

Q4.2: P3, P2, P4, P1

SJF chooses the shortest processing time first: 10, 15, 20, 25.

Q4.3: 20 ms

SJF waits are 0, 10, 25, 45; average = $80/4$.

Q5.1: 2 ms

$0.5 \text{ MB} / 250 \text{ MB/s} = 0.002 \text{ s} = 2 \text{ ms}$.

Q5.2: 9 ms

Rotation period at 10000 rpm is 6 ms; average rotational latency is 3 ms. $TIO = 4 + 3 + 2$.

Q5.3: 250,000 I/O operations

$125 \text{ GB} = 125,000 \text{ MB}$; $125,000 / 0.5 = 250,000$ chunks.

Task Parallelism

Q1.1: 3

After T1 completes, T2, T3, and T4 can run independently.

Q1.2: 45 time units

The longest path is $T1+T3+T6+T7 = 10+20+10+5$.

Q1.3: $85/45 = 17/9 = 1.89x$

Serial work = $10+15+20+15+10+10+5 = 85$. Parallel time = 45.

Q1.4: 30 time units

In an optimal 3-worker schedule, the worker that runs only T4 works 15 units and is idle 30 units.

Q1.5: 50 time units

With 2 workers, all predecessors of T7 can finish by time 45; T7 then adds 5.

Q2.1: 60 time units

For $x = 25$, work = 80 and the 4-worker optimum is 35, so total idle = $4 \cdot 35 - 80$.

Q2.2: 35 time units

For $x = 25$, the longest paths are T1+T5+T6 and T3+T6, both 35.

Q2.3: 40 time units

For $x = 10$ with 2 workers, all predecessors of T6 can finish by time 30; T6 then adds 10.

Data Parallelism**Amdahl's law: $10/3 = 3.33x$**

With parallel fraction $p = 0.70$ and infinitely many processors, speedup = $1 / (1 - p) = 1/0.30$.